

Nonlinear Radial Correlation of Electrostatic Fluctuations in the STOR-M Tokamak

W. X. Ding, C. Xiao, D. White, M. Elia, and A. Hirose

Plasma Physics Laboratory, University of Saskatchewan, Saskatoon, Saskatchewan, Canada S7N 5E2

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A practical algorithm to calculate the nonlinear correlation is developed. This algorithm permits the determination of not only the correlation strength but also the correlation direction. Application of the algorithm to data obtained in the edge plasma in the STOR-M tokamak with Langmuir probes indicates that the direction of fluctuation correlation reverses after L - H transition and that the nonlinear correlation length is 4 times as long as the linear correlation length. [S0031-9007(97)04164-1]

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The role played by plasma turbulence in anomalous transport and L - H transition [1] continues to be a central topic in research of magnetically confined high-temperature plasmas. In characterization of plasma fluctuations, the method based on correlation functions is widely used. However, the conventional correlation method has some difficulties. First, in a complex nonlinear system (e.g., brain, fusion plasmas), linear independence between two variables does not mean that two variables (or subsystems) are independent [2] unless their probability density distribution is Gaussian. Plasma turbulence observed in high density plasma exhibits non-Gaussian statistics [3]. Second, a global mode [4] (self-organized criticality mode [5]) is proposed to explain the transport dynamics in a tokamak, and there is some experimental evidence [6,7] to support the existence of the global mode. However, the correlation length of local fluctuations estimated from the conventional correlation technique is much smaller than that expected. Third, the standard correlator requires statistical averaging over a long time period which is sometimes difficult because of changes in discharge parameters, and it is unable to reveal a causality relation. To quantify the long range coupling of turbulent cells when the coupling becomes weak and determine the direction of correlation, we employ a new correlation technique and analyze turbulence data in the STOR-M tokamak during L - H mode transition. The algorithm of the new correlation technique [8] has been tested numerically for a system of two coupled Van der Pol oscillators. The algorithm allows us to determine not only the correlation strength but also the correlation direction. The algorithm is applied to edge potential fluctuation data observed in L - H transition in the STOR-M tokamak. The direction of fluctuation correlation is outward for L mode and inward for H mode. A long radial correlation length is found.

The algorithm is based on reconstructing vector dynamics [9] from scalar time series. We consider two variables $x(t)$ and $y(t)$ in a complex system (e.g., potential fluctuations), and reconstruct a finite dimension (M) phase space from data $x(i\Delta t), y(i\Delta t), i = 1, 2, \dots, N$, N data length; the constructed M -dimensional points $\mathbf{x}_i^M =$

$M^{-1/2}(x_i, x_{i+m}, \dots, x_{i+(M-1)m})$, $\mathbf{y}_i^M = M^{-1/2}(y_i, y_{i+m}, \dots, y_{i+(M-1)m})$ completely describe system states in the phase space for sufficiently large embedding dimension M , where Δt is the sampling time interval and $\tau = m\Delta t$ is a time delay. As a first order approximation, τ is chosen as the delay time at which the autocorrelation function of time series falls to e^{-1} . The conditional probability $P(y_i^M | x_i^M)$ formulated by Packard *et al.* [10] is the probability to find y_i^M for fixed x_i^M . If they belong to independent subsystems, the conditional probability would not depend on x_i^M and depend on the distribution of y_i^M alone. Therefore, when the distance $|x_i^M - x_j^M| < \epsilon$ (small quantity compared to the size of attractor $\epsilon_{\max}$), one expects that $|y_i^M - y_j^M| \approx \epsilon$ for correlated x and y , where $|y_i^M - y_j^M|$ is approximate to the mean distance between points y_i^M for independent x and y . Here $||$ is the Euclidean distance. This ϵ interrelation of the x_i^M and y_i^M can be estimated by the mean conditional dispersion [8]:

$$\sigma_{xy}^M(\epsilon) = \sqrt{\frac{\sum_{i \neq j} |y_i^M - y_j^M|^2 H(\epsilon - |x_i^M - x_j^M|)}{\sum_{i \neq j} H(\epsilon - |x_i^M - x_j^M|)}}, \quad (1)$$

where $H(x)$ is the Heaviside function. The correlation coefficient is defined as $g_{xy} = \lim_{M \rightarrow \infty} \epsilon_0 / \epsilon_{\max}$, where $\sigma(\epsilon_0) = 0.9 \times \sigma_{\max}$, $\epsilon \in (\epsilon_{\min}, \epsilon_{\max})$, $\epsilon_{\min}(\epsilon_{\max})$ is the minimum (maximum) size of the attractor. It is reasonable to assume that two variables are correlated when $\epsilon \geq 10\epsilon_{\min}$. Asymmetry $g_{xy} \neq g_{yx}$ reflects the direction of correlation. We define $s_{xy} = -(g_{yx} - g_{xy}) / (g_{xy} + g_{yx})$ as a measure of asymmetry. $s_{xy} > 0$ indicates that the direction of correlation is from x to y . The normalized correlation coefficient can be defined as $G_{xy} = g_{xy} / \sqrt{g_{xx}g_{yy}}$. Finally we recall the standard correlator defined as $C_{xy} = \langle xy \rangle / \sqrt{[\langle x^2 \rangle \langle y^2 \rangle]}$, where $\langle \rangle$ denotes the time average, and $C_{xy} = C_{yx}$. We apply the algorithm to the dynamical data generated from the coupled Van der Pol oscillators which exhibit undriven chaos [11]. The system has been used to model some plasma problems and its nonlinear dynamics has been studied intensively [12]. The model equations

are

$$\frac{d^2x}{dt^2} = [a_1 - (x + \beta y)^2] \frac{dx}{dt} - (x + \beta y), \quad (2)$$

$$\frac{d^2y}{dt^2} = [a_2 - (y + \alpha x)^2] \frac{dy}{dt} - (y + \alpha x), \quad (3)$$

where α and β are coupling coefficients between two oscillators. The equations are integrated numerically by a standard Runge-Kutta method with a time step 0.01. We identified eleven chaotic attractors from the system. Their control parameters are $\beta = -3, -3.25, -3.55, -3.75, -4$ for $\alpha = 0.75, \alpha = 1.75, 2.0, 2.25, 2.5, 3.0$ for $\beta = -0.5$, and $\alpha = 0.75, \beta = -0.75$, respectively ($a_1 = a_2 = 1$). Those chaotic attractors have been recognized in Ref. [11] and are confirmed by calculating the largest Lyapunov exponents [13]. The algorithm has been applied to the simulated data ($\alpha = 2.5, \beta = -0.5$) (Fig. 1). Nonlinear correlation coefficient g between two oscillators (2) and (3) can be obtained from intersection of the dispersion curve ($M = 9$) and the straight line $\sigma = 0.9$.

The numerical test shows that g_{xy} and g_{yx} saturate with the increase in embedding dimension. The case shown in Fig. 1 indicates that the influence of subsystem y on the subsystem x is larger than that of x on y . It means that oscillator (2) drives oscillator (3). Numerical results confirm that $g_{xy} > g_{yx}$ for $|\alpha| > |\beta|$, $g_{xy} < g_{yx}$ for $|\alpha| < |\beta|$, $g_{xy} = g_{yx}$ for $|\alpha| = |\beta|$. The ability to show the master-slave relation is a new property of the nonlinear correlator. We have applied both correlators to the simulated data to reveal the difference between them. The result is shown in Fig. 2(a) where both correlation coefficients are normalized to the value at $\alpha = 1.75$ in order to compare them quantitatively. Both correlation coefficients decrease with the increase in control parameter α . The linear correlation decreases with α faster than the nonlinear correlation. The difference between the two correlators becomes more pronounced when the coupling between two oscillators becomes weaker. This supports the view that the lack of a linear correlation does not mean the independence of two variables [2].

In order to apply the algorithm to real experimental data which contain unavoidable noise we have studied the influence of noise on the calculation of the nonlinear

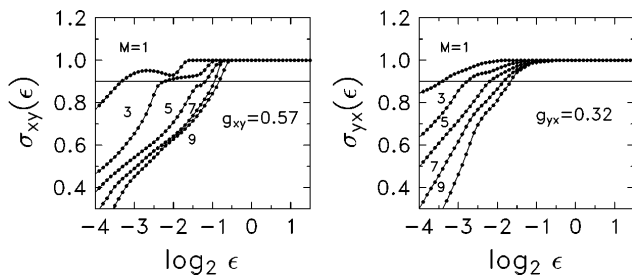


FIG. 1. Conditional dispersion curves are calculated based on simulated data from Eqs. (2) and (3) for $\alpha = 2.5, \beta = -0.5, N = 1000$ without noise.

correlator. A Gaussian noise is added to time series $x(t), y(t)$ generated by coupled Van der Pol equations. The nonlinear correlation coefficient slightly decreases with amplitude of noise up to 10% as shown in Fig. 2(b). The noise reduces the correlation coefficient as expected. Therefore the algorithm is applicable for experimental data which may include a noise level less than 5%.

To gain insight into the interplay of fluctuations and transport in the STOR-M tokamak, we have investigated the radial correlation of electrostatic fluctuations in L mode and H mode with the new algorithm. Experiments are performed in STOR-M tokamak which has a major radius of $R = 46$ cm and a minor radius of $a = 12$ cm. Normal Ohmic hydrogen discharge parameters are toroidal magnetic field $B_T \approx 0.7$ T, plasma current $I_p = 20$ kA for 25 ms, loop voltage $V_l = 2-4$ V, average electron density $\bar{n}_e = 1.0 \times 10^{19} \text{ m}^{-3}$, average electron temperature $\bar{T}_e = 200$ eV, and global energy confinement time $\tau_E \approx 1$ ms. Two identical sets of Langmuir probe, which are radially separated $\Delta_r = r_2 - r_1$ cm, are inserted from the outer board of the tokamak in the midplane. Each set consists of three tips. Fluctuations of potential can be measured from floating potential ($\tilde{V}_\theta$) of two tips poloidally separated by $\Delta_\theta = 2$ mm and the poloidal electric field can be deduced from $\tilde{E}_\theta = (\tilde{V}_{\theta_1} - \tilde{V}_{\theta_2})/\Delta_\theta$. The other tip is used to measure the ion saturation current. Therefore the potential fluctuation $\tilde{\Phi}$, poloidal electric field fluctuation $\tilde{E}_\theta$, and plasma density fluctuation are acquired simultaneously in a shot by a digitizer (8 bit, sample rate 2 MHz, data length 16 kb), and time average fluctuation-induced flux $\langle \Gamma \rangle = \langle \tilde{n}_e \times \tilde{E}_\theta \rangle / B_T$ can be obtained. Figure 3 shows the time evolution of the discharge. H mode is triggered by applying negative bias voltage on an electrode (diameter 2 cm) located at $r = 0.8a$. Distinct sharp reduction in the H_α radiation accompanied by an increase in the plasma density is an indication of Ohmic H mode. Plasma potential fluctuations (v_1, v_2) at different radial locations, corresponding

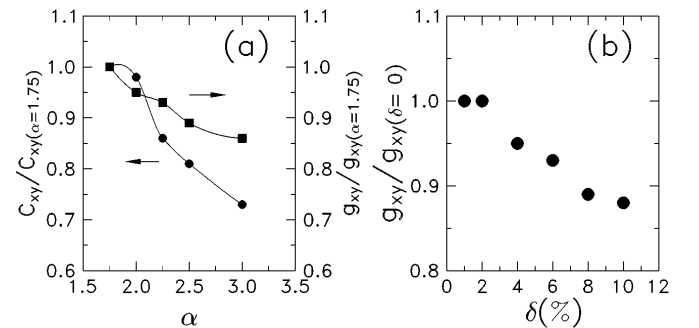


FIG. 2. (a) Comparison between the nonlinear correlator and linear correlator for $\beta = -0.5, N = 5000$ for the linear correlator, $N = 2000$ for the nonlinear correlator. (b) The nonlinear correlation coefficient (g_{xy}) versus the amplitude of noise. Generated data come from $\alpha = 3.0, \beta = -0.5, N = 2000$.

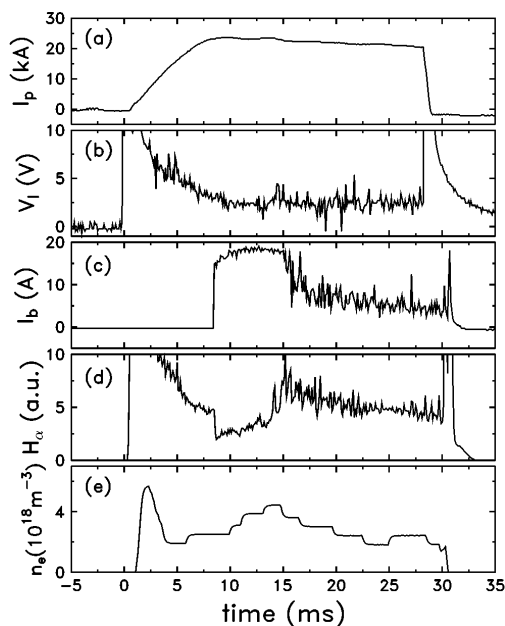


FIG. 3. Discharge history with biasing voltage -433 V ; (a) plasma current; (b) loop voltage; (c) biasing current; (d) H_α radiation; (e) electron density. Shot No. 16484.

to power spectra and their conditional dispersion curves during normal discharge without bias are shown in Fig. 4. Two positive Lyapunov exponents for both attractors indicate that two attractors are hyperchaotic [13]. L -mode data are acquired without bias. H -mode data are acquired when H_α has a sharp reduction [Fig. 5(d)]. Compared with the L -mode, the H -mode phase has a suppression of potential fluctuations (a reduction of 30% to 50%) and a low frequency peak $f_0 = 25 \text{ kHz}$ and their harmonic merge from a broad turbulent spectrum. The electric field $\sqrt{\langle E_\theta^2 \rangle}$ is 10 V/cm for H mode and 20 V/cm for L mode.

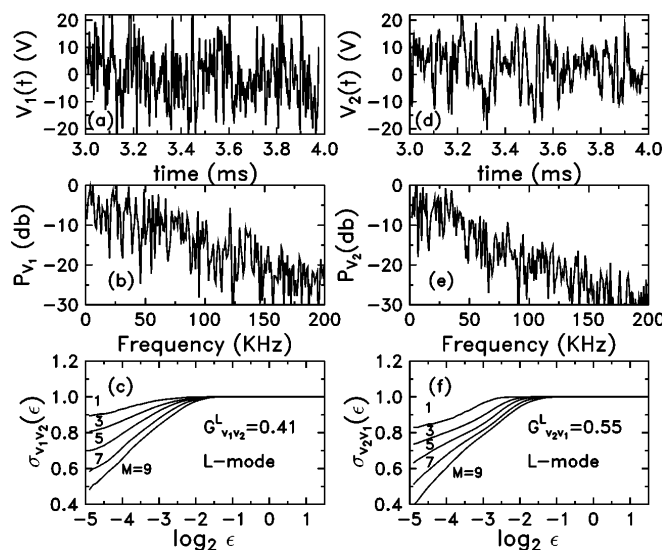


FIG. 4. L -mode experimental results. v_1 is the plasma potential at $r_1 = 0.83a$, v_2 at $r_2 = 0.95a$. Shot No. 16542.

Correlation direction is changed after L - H transition, i.e., $s^L_{v_1v_2} = 0.16 \pm 0.06 > 0$, $s^H_{v_1v_2} = -0.11 \pm 0.08 < 0$, where superscript L (H) denotes L mode (H mode). Error bars are carefully estimated from different shots, the length of data section, and the time lag τ . $s^L_{v_1v_2} > 0$ indicates that the correlation direction is from r_1 to r_2 in L mode. In other words the probability of finding the inner oscillations at the outer is larger than the probability of finding the outer oscillations at the inner for L mode. The nonlinear correlation coefficients G^L for L mode and G^H for H mode are shown in Figs. 4(c) and 4(f) and Figs. 5(c) and 5(f), respectively. To understand the correlation direction physically we use two poloidally separated probes to investigate poloidal correlation. In the edge region of plasma, the correlation direction is in the electron diamagnetic drift direction, and in the scrape-off layer, the correlation direction is in the ion diamagnetic drift direction, which are consistent with the previous observation based on phase shift of fluctuation signals [14]. The correlation direction also reflects the direction of fluctuation propagation. Using the same algorithm mentioned here, one can also judge the direction of a wave train, which agrees with the result of direct measurement [15]. Therefore the propagation direction of oscillation is outward in L mode. $s^H_{v_1v_2} < 0$ for H mode indicates that correlation direction is from r_2 to r_1 and the propagation direction is inward.

The radial correlation length of potential fluctuations has been measured in L mode and H mode, respectively. Spatial correlation can be estimated either by a conventional correlator or by nonlinear correlator as shown in Fig. 6. Nonlinear correlation length ($l_n \approx 2 \text{ cm}$) is 4 times as long as linear correlation length ($l_l \approx 0.5 \text{ cm}$). The suppression of edge turbulence [16] and the radial decorrelation [17] of fluctuation are generally considered

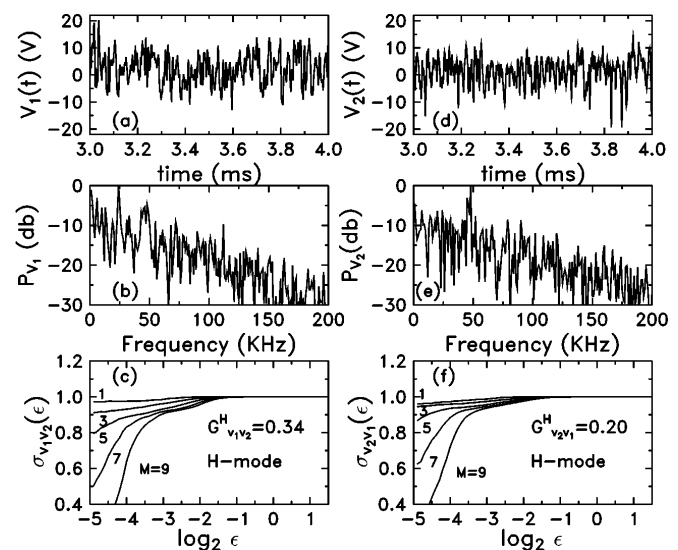


FIG. 5. H -mode experimental results. v_1 is the plasma potential at $r_1 = 0.83a$, v_2 at $r_2 = 0.95a$. Shot No. 16484.

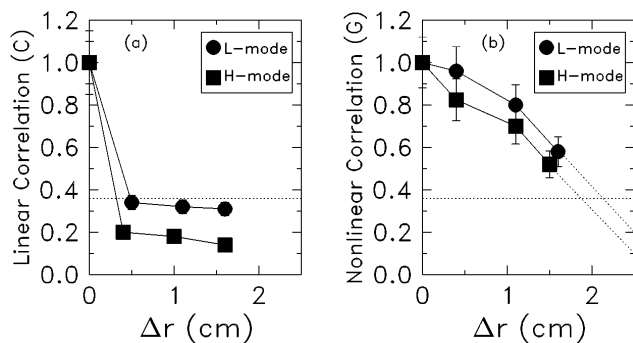


FIG. 6. Spatial correlation length is defined as the distance at which the normalized correlation coefficient decreases to $1/e$; (a) linear correlator; (b) nonlinear correlator. A probe is located at $r_2 = 12$ cm.

to be responsible for the improvement of confinement in fusion devices. Recent experimental studies have shown that the edge fluctuation can be suppressed by applying radial electric field in tokamaks [14,18,19]. The radial decorrelation of turbulent cells is required by the local-mode theory in order to interpret the reduction of transport in a narrow region near plasma edge. But the well extended radial correlation of local turbulent cells is required by the nonlocal-mode theory to interpret simultaneous drop of transport coefficients over a larger range in L - H transition [7]. Therefore the measurement of correlation length is important to interpret transport properties in L - H transition. Plasma turbulence in a tokamak may form a chain of convective cells across plasma minor radius because of tokamak's toroidal geometry [4]. We have observed a longer radial correlation length, which may support the global model.

Diffusivity can be estimated from nonlinear correlation length. The diffusive decorrelation time is $\tau_D = l_n^2/D^L = l_n^2 \nabla n_e / \langle \Gamma \rangle \approx 10 \mu s$ for L mode [Fig. 4(a)] and $30 \mu s$ for H mode [Fig. 5(a)], where diffusion coefficient D is determined from the locally measured particle flux $\langle \Gamma \rangle$ and the density gradient. The decorrelation time estimated from the nonlinear correlation length and diffusivity is consistent with the nonlinear correlation time τ_N at which nonlinear autocorrelation function $g_{x(t)x(t+\tau_N)}$ falls to e^{-1} . On the other hand diffusion comes from the randomization both in space and time based on nonlinear dynamics. Intrinsic relations among transport properties, positive Lyapunov exponents, and the Kolmogorov-Sinai (KS) entropy for different systems have been established [20]. To estimate diffusivity we employ a simple relationship between diffusion and the entropy of system $D = L^2 \cdot S$ ($S = \sum_{\lambda_i > 0} \lambda_i$) as suggested in Ref. [20] in our system [21], where L has a length dimension and λ_i (with an entropy dimension, sec^{-1}) is Lyapunov exponent spectra [13] calculated from time series. From experimental data we have $L = \sqrt{D/\sum \lambda} = 3$ cm for L mode, which is the order of magnitude of the nonlinear correlation length. Therefore the nonlinear correlation length we

present in this paper connects diffusion and the entropy of plasma system. Furthermore let us calculate the diffusion coefficient in H mode: $D^H = (l_n^H/l_n^L)^2 (S^H/S^L) D^L = (1.85/2.14)^2 (40\,000/140\,000) D^L = 0.2 \times D^L$. This also agrees with experimental results $D^H \approx 0.25 \times D^L$.

In conclusion we introduce a new tool to explore the nonlinear radial correlation between fluctuations during L - H mode transition in a tokamak. The approach may open a possibility to find a large scale ($\sim a$) correlation in a tokamak.

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